

SHOGUN design pack (2026-10-06)

Working name SHOGUN. A fan instrument in the Jidai Collection. Design pack and voice engine in this commit.

SHOGUN is a drum computer and two note voices, specified from the first MFB Tanzbär: the 17-page English user manual, the matching German Bedienungsanleitung, and that instrument's MIDI control-change list. The jobs of the voices follow those sources and the design brief. The panel, when it is drawn, is a new face. This commit has the design pack and the voice engine in engine/. Panel art is a later commit.

Revisions

- 2026-10-06: first pack. Sources, clock bypass, voice equations, sequencer, tests, and the PDF of the same text.
- 2026-10-06: voice engine. Framework-free C++ in engine/, INT and EXT trigger paths, and the named tests in tests/voices.cpp.

What this is

A stand-in design. Every coefficient in SCHEMATICS.md is marked STAND-IN. Where the English manual, the German manual, and the control-change list disagree, both wordings are quoted and the pick is marked ASSUMED or CHOICE. No service schematic was in the sources. The equations are functional blocks, not a transistor circuit.

The instrument this pack describes:

- 14 drum tracks and 2 note tracks
- An internal sequencer with patterns of 1 to 32 steps
- A clock switch, INT or EXT, that connects or disconnects that sequencer from the voice triggers
- Paired outputs plus a main mix

Files

File	Role
README.md	This front page
METHODOLOGY.md	Sources, quotations, and the choices
SCHEMATICS.md	Block, sample equation, knob range, and a 48 kHz example per voice
BUILD_GUIDE.md	Engine layout, PDF build, and the panel rule
TESTPLAN.md	Named tests and the numbers they lock
REPO_SETUP.md	Tree, git, and the collection link
engine/	Voice engine, clock, pattern, and outs. No JUCE
tests/voices.cpp	The named checks, compiled by the Makefile
Makefile	Builds build/shogun_tests
docs/SHOGUN_Design_Pack.pdf	The same documents in one PDF

Rebuild the PDF with:

```
python3 scripts/build_design_pack_pdf.py
```

The Jidai Collection

BUSHIDO and RONIN share one patch format and one cable feel. JIDAI RACK compiles those two engines. SHOGUN's engine is in this repository. It is not compiled into that rack.

- **BUSHIDO**: a 3 x 12 analog step sequencer with real patch cables. <https://github.com/martialsystems/bushido>
- **RONIN**: a semi-modular synthesizer you patch as an effect. <https://github.com/martialsystems/Ronin>

- **JIDAI RACK:** BUSHIDO and RONIN in one rack, in one plugin. <https://github.com/martialsystems/jidai-collection>

Legal

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Methodology (2026-10-06)

Status

This is a stand-in design for a fan instrument. The voices are analog blocks: oscillators, envelopes, noise, and a clip. Hats and cymbals are square or metallic stacks plus noise. No sample is in the voice model. No coefficient is a measured part value. Each one is marked STAND-IN in SCHEMATICS.md.

No service schematic was found in public, and none is invented here. A block diagram is the farthest the circuit claim goes.

Sources

Read on 2026-10-06.

1. German primary. *MFB Tanzbär Bedienungsanleitung*, 17 PDF pages, two printed pages on most sheets, printed page numbers through 31. PDF metadata: CreationDate D:20131024121806+02'00, Creator Adobe InDesign CS5 (7.0), Producer Adobe PDF Library 9.9. Text extracted from the file published at https://media.djmania.net/manuales/pdf/Manual_MFB_Tanzbar.pdf. The MIDI appendix is printed page 31 of this file. That appendix is the control-change list used below.

2. English primary. *MFB Tanzbär User Manual*, the 17-page English edition with the same contents list (overview through MIDI implementation). Printed pages quoted below were read from <https://www.manualshelf.com/manual/mfb/tanzb-ar/user-manual-english/page-3.html> and the neighbouring pages of that edition (pages 4, 6, 8, 9, 11). A separate English PDF file was not downloaded. The contents list and the instrument chapter match the German edition's structure.

3. Published control-change republication. <https://midi.guide/d/mfb/tanzbar-1/>, last update 12 September 2024, 64 parameters. This is the list identified by BD1 Attack on CC 2, Decay on 64, Pitch on 65, Tune on 3, Noise on 4, Filter on 5, Dist on 6, and Trigger on 66. Where it disagrees with the 2013 appendix, the appendix wins and the row is marked.

A secondary web conversion at <https://manuals.plus/mfb/mfb-tanzbar-analog-drum-machine-manual> was read only for the Data-versus-Sound conflict named in the brief. It is not a primary.

Tanzbär 2, Tanzmaus, and Tanzbär Lite were not used.

Marks

- STAND-IN: a number chosen so the equation is definite. It is not a capture.
- CHOICE: the sources, or the sources plus the brief, support this reading, and the pack uses it.
- ASSUMED: the sources disagree, or they are silent, and a single reading is still required.

Parameter names

Both primary instrument chapters agree on bass drum 1.

English, printed pages 14 to 15: Attack is the level of the attack transients. Decay is volume decay time. Pitch is time and modulation intensity of the pitch envelope. Tune is pitch. Noise is noise level. Filter is the sound of the noise signal. Data is distortion level (printed "Distorion"). Sound selects 1 of 16 different attack transients.

German, printed pages 14 to 15: Data "regelt die Intensität des Verzerrers". Sound: "16 verschiedene Attack-Transienten stehen zur Auswahl". Filter "beeinflusst den Klang des Rauschanteils".

The 2013 appendix names the same jobs as BD1_DIST on CC 6 and BD1_TRIGGER on CC 66. It has no row named Sound.

The secondary conversion says BD1 Data is "Sound of noise signal". That sentence swaps Data with the job both primaries give to Filter.

CHOICE: Sound, which the appendix calls BD1_TRIGGER, selects the transient, index 0 to 15. Data, BD1_DIST, is the clip. Filter stays on the noise.

Quoted disputes

Individual outputs, overview against the jack list

English overview, printed page 4: "Individual outs (in pairs except claps)."

German overview, printed page 4: "Einzelausgänge (bis auf Bassdrum und Clap paarweise ausgeführt)."

Both rear-panel lists then say: BD left is bass drum 1 and BD right is bass drum 2. SD/RS is snare and rim. HH/CY left is open and closed hat, right is cymbal. The clap out is stereo, with the attack transients spread in the field. TO/CO spreads the three toms or congas. CB/CL left is clave, right is cowbell.

The two overview sentences disagree with each other. The jack lists agree with each other and with the brief.

CHOICE: follow the jack list. Maracas, the lead voice, and the bass voice are not on that list. They go to the main mix only.

Cable in an individual out

English printed page 7: "If you plug in a cable on an instrument out, the sound is erased from the main out."

German printed page 7: "Wird ein Kabel in einen Einzelausgang gesteckt, werden die betreffenden Sounds im Hauptausgang gelöscht."

The two primaries agree: the hardware removes those sounds from the main output.

CHOICE for SHOGUN: an individual out is a tap. The voice stays in the main mix. The hardware sentence is recorded so the difference is explicit. TESTPLAN.md locks it as `testIndividualOutStaysInMix`.

Seventeen names, fourteen tracks

Both overviews say 17 drum instruments and 14 tracks that trigger drum instruments. The instrument chapter lists 14 drum names: BD1, BD2, SD, RS, CY, OH, HH, CL, CP, LTC, MTC, HTC, CB, MA. The two note tracks are extra.

ASSUMED: each tom track has a tom mode and a conga mode, and the overview counts those as six names on three tracks. 11 other drums plus 6 is 17 names on 14 tracks. The manuals do not write that sum.

Tom and conga buttons

English, printed pages 15 to 16, gives LTC, MTC, and HTC the same buttons: step button 12 toggles tom and conga, step button 13 enables noise.

German, printed pages 15 to 16: LTC uses the lit LEDs 12 and 13, MTC uses 13 and 14, HTC uses 14 and 15.

The appendix gives each voice its own noise switch and its own tom/conga switch, plus one shared TOM_NOISE on CC 84.

ASSUMED: the voice model uses the per-voice switches and the shared level. The button numbers are a panel procedure, and the two editions do not agree, so they are not part of the equation.

Track length

German overview: "Länge der Spur (1 bis 32 Steps)."

German record chapter, printed page 23: each track may have an individual length "zwischen einem und 16 Steps".

English overview: step number 1 to 32, with four scale settings.

CHOICE: pattern length and per-track length are integers from 1 to 32, which is the overview and the brief. Steps 1 to 16 are the A half. Steps 17 to 32 are the B half. The "1 to 16" sentence matches the sixteen step buttons on the A half.

Scale and length

Both manuals tie scale to a default pattern length: 32nd notes give 32 steps, 16th-note triplets give 24, 16ths give 16, 8th-note triplets give 12. Measure can then set another count. The bar still takes the same time, so a 32nd scale runs the sequencer twice as fast as a 16th scale.

The brief asks for scale as step length, and for patterns of 1 to 32 steps as their own setting.

CHOICE: in SHOGUN, scale sets only how many steps fit in a quarter note. Pattern length is a separate integer from 1 to 32. The hardware coupling is not copied. This split is ASSUMED relative to a machine whose scale button also picked the length.

Cowbell resolution

Both instrument chapters: Data offers 16 tunings, and Sound is the decay time.

Appendix: CB_Tune on CC 85, value 0 to 127, and CB_Decay on CC 86, value 0 to 127.

ASSUMED: Tune and Decay are continuous 0 to 127, under the appendix names. A 16-step control is a quantizer of Tune, step i mapping to CC $\text{round}(i \times 127 / 15)$. The brief's two detuned squares are the oscillator. The manuals do not name the waveform.

Clap type and count

Both instrument chapters: Decay is the tail, Filter is colour, Attack is the level of the attack transients, Data is the number of those transients, Sound is 16 different transients.

The record chapter's hidden-parameter paragraph names BD1, the toms and congas, and the cowbell. It does not name the clap. English printed page 20 and German printed page 20 both still say flam cannot be programmed on Clap, CV1, or CV2/3.

Appendix: CP_DECAY 75, CP_FILTER 18, CP_ATTACK 76, CP_TRIGGER 77. No row is named count, and no row is named Sound.

ASSUMED: CP_TRIGGER is the type, on the same pattern as BD1_TRIGGER. The count is Data, an integer from 1 to 8, and it has no published CC. The record paragraph's shorter instrument list is treated as incomplete next to the instrument chapter.

Set select and one track-delay label

2013 appendix: Set Select, controller 0, value 0 to 2.

midi.guide: Set Select has no controller number, and the range column says 0 to 127.

2013 appendix: Track Delay HH, controller 97.

midi.guide labels controller 97 "Track Track HH". The number matches.

CHOICE: controller 0 with values 0, 1, and 2, and the name Track Delay HH for 97.

Clock bypass

INT and EXT are SHOGUN. They are not a switch in the Tanzbär manuals. Those manuals have MIDI clock, an analog sync jack that can be an input or an output, and a manual-trigger mode. None of those is this switch.

The switch is the only control that selects the master.

- INT: the internal sequencer triggers the voices. Its tempo is the host tempo while the host is playing, otherwise its own tempo. A cable in a Trig jack does not fire the voice.
- EXT: the internal sequencer does not trigger the voices. Step on and off, flam, and accent from the pattern are ignored for triggering. Knob values still apply. Each voice fires from its own Trig jack. Accent may arrive as velocity on that jack.
- EXT does not stop the clock. The step counter and the clock display keep advancing. Switching back to INT connects the triggers to the step the clock has reached.
- A cable in a Trig jack does not force EXT.

ASSUMED: "resumes" means the counter never froze. The alternative, a frozen counter that continues from the step where EXT was engaged, would stop the clock the brief says keeps running.

Voices the manuals do name

The instrument chapter gives the jobs below. Ranges and waveforms that the chapter does not give are in SCHEMATICS.md, marked there.

- BD1: attack level, decay, pitch envelope time and depth on one knob, tune, noise level, noise filter, distortion, and 1 of 16 attack transients.
- BD2: decay up to a steady tone, tune, and tone as the attack level. The chapter does not give BD2 a noise, a filter, a distortion, or 16 transient types. It is a second kick.
- SD: tune shared by two tones, detune of tone 2, snappy noise level, noise decay, blend of the two tones, decay of both tones, and pitch-envelope intensity.
- RS: pitch only. The brief asks for a short pitched click.
- CY: decay, blend of both signals, and pitch or colour.
- OH and HH: separate decays, and one pitch or colour shared by both. The appendix exposes that colour as HH_TUNE on CC 73. There is no OH tune row.
- CL: tune and decay.
- CP: tail decay, filter, attack level, a count of attack transients, and 16 transient types. The clap out spreads those transients. The brief adds the spacing, about 10 to 12 ms, which the manuals do not print.
- LTC, MTC, HTC: tune, decay up to a steady tone, noise on or off, tom or conga, and one noise level for all three. Bend is available. The tom out spreads the three.
- CB: tune and decay in the appendix. The brief specifies two detuned squares.
- MA: decay only. The brief specifies filtered noise.
- Lead and bass: the overview says one parameter each. Bass Data is "Filter cutoff or CV 3 value" in the English chapter. Setup plays a steady 440 Hz and tunes both voices with Data. The waveform is not named. ASSUMED waveform: a band-limited saw for each. ASSUMED lead parameter: a one-pole tone, because the chapter names one parameter and does not name the lead knob.

Closed hat kills open hat. The pages read do not state a choke time. The brief requires the choke, so the pack defines it: the closed-hat trigger clears the open-hat envelope on that sample.

Sequencer the manuals do name

- Patterns live in 3 sets of 3 banks of 16, which is 144. That memory map is the hardware's. SHOGUN stores patterns without a 144-slot cap in this pack. The set-select controller is still defined.
- A and B. A pattern longer than 16 steps has a B half, and the machine walks it. A/B can also alternate. Copy of A onto B exists on the hardware. Chain is runtime only: the English overview calls it a fill that is not storable, and the German overview says the chain is "nicht speicherbar".
- Per track: length, shuffle, shift, mute. Shuffle in record mode is per track, 16 intensities. Shuffle in play mode is global and does not replace the stored per-track values when it is released. Shift is controllers 89 to 104, described as a small delay of the whole track. No millisecond range is printed.
- Per step, drums: on or off, one of three accent levels (off or soft, green or medium, red or loud), flam on 16 patterns, bend where the instrument has it, and an extra sound value where the instrument chapter has one. Bend instruments are BD1, BD2, SD, LTC, MTC, and HTC. Flam is unavailable on clap and on both note tracks.
- Per step, note tracks: on or off, pitch over three octaves, accent on the bass track, and a second value on the bass track that is CV3 and the internal bass cutoff. A/B enters a silent step. Select ties steps into a longer note.
- Level pots and the master volume are not stored in the pattern.
- Manual record mode shares one knob set across steps. Accents and flams can still differ per step. Step and jam modes can store different knob values per step.

Tempo

English play chapter: the tempo knob covers approximately 60 to 180 BPM, and there is no tempo readout.

CHOICE: SHOGUN's internal tempo uses those endpoints as exact bounds, default 120. The clock display is part of SHOGUN. The hardware has no readout.

Gate level

A Trig jack is a rising edge through 1.0 V on a gate that rests at 0 V and is high at 5 V. That 1 V threshold is the collection's existing gate rule between RONIN and BUSHIDO. Velocity, when the sender has one, is an integer 0 to 127 riding with the edge. A plain gate with no velocity uses 127.

ASSUMED: the plain-gate default of 127. The Tanzbär manuals say drum notes carry velocity. They do not define a jack that has no velocity byte.

Schematics (2026-10-06)

Functional blocks and sample equations. Coefficients are STAND-IN. Sample rate in every numeric example: 48,000 Hz. Arithmetic is IEEE-754 double. A later test may treat the printed decimals as exact within 1e-5.

Shared helpers:

```
u(cc) = cc / 127
fs = 48000
pi = 3.141592653589793
```

cc is an integer from 0 to 127. A per-step override, when the record mode stored one, replaces the live knob for that step. Manual record mode does not store per-step knobs.

A rising trigger at sample n0 resets that voice's phase to 0 and restarts its envelopes at n0. Until the first trigger, a cleared voice outputs 0.

Velocity law

Pattern accent index a is 0, 1, or 2. English printed page 20: LED off is soft, green is medium, red is loud.

```
g_accent(0) = 0.55
g_accent(1) = 0.78
g_accent(2) = 1.00
```

STAND-IN.

EXT velocity v , an integer 0 to 127:

```
g_vel(v) = 0.15 + 0.85 * (v / 127)
```

STAND-IN. $v = 0$ gives 0.150000. $v = 100$ gives 0.819291. $v = 127$ gives 1.000000.

INT multiplies the voice by g_accent and ignores Trig velocity. EXT multiplies by g_vel and ignores the pattern accent. The gain multiplies the voice output after the voice equation.

Clock

Steps per quarter note, from the four scale settings in both manuals:

Scale	Steps per quarter	Samples per step at 120 BPM
32nd	8	3,000
16th triplet	6	4,000
16th	4	6,000
8th triplet	3	8,000

```
period = fs * 60 / (bpm * steps_per_quarter)
```

At 120 BPM and a 16th scale, period is 6,000 samples. Internal bpm is clamped to 60 through 180. Default 120.

The step counter increments on each period boundary in INT and in EXT. The clock display shows $(\text{counter} \bmod \text{pattern_length}) + 1$. pattern_length is the bar, an integer from 1 to 32.

Each track stores its own length L , also 1 to 32. On global counter c the track reads step $c \bmod L$. A shorter track cycles against the bar. A longer track has steps past one bar, which is how a track reaches the B half when the bar is shorter. The track fires when that stored step is on. It does not also have to fall inside the bar length.

Shuffle intensity s is an integer 0 to 15. STAND-IN, even steps only (0-based step index odd):

```
delay_samples = (s / 15) * (period / 3)
```

$s = 0$ is no delay. $s = 8$ at period 6,000 delays the odd step by 1,066.666667 samples. The fractional sample is kept and the trigger fires on the sample that contains the fractional boundary. Play-mode global shuffle, while held, uses one s for every track and does not write the pattern.

Track shift, controllers 89 to 104, one per track including the two note tracks. STAND-IN:

```
shift_samples = round(u(cc) * 0.030 * fs)
```

Full scale is 1,440 samples, 30 ms. The whole track's triggers move later by that many samples.

INT and EXT

INT fires a drum voice when its step is on. The flam table and the accent index apply. The Trig jack is not read.

EXT does not fire from the pattern. Flam, step on and off, and pattern accent do nothing. The Trig jack's rising edge fires the voice, and g_vel supplies accent. Knobs, including per-step knob overrides if a later editor still wants them, remain in force. This pack's EXT path uses the live knobs, not the pattern's per-step sound overrides, because the brief says the pattern is ignored for step on and off, flam, and accent, and that knob values still apply. ASSUMED: "knob values" means the live knobs, not values stored on ignored steps.

Bend stored on a step is part of the pattern. EXT ignores it. The Pitch knob on BD1, BD2, SD, and the toms still applies, because that knob is a sound control, not the step bend.

Noise generator

One generator for the whole instrument, so examples reproduce.

```
state = (1664525 * state + 1013904223) mod 2^32
x = state / 2^32 * 2 - 1
```

Initial state is 1. The first draw yields $x = -0.527089$. Each noise consumer takes the next draw when it needs a sample. The maracas example below is the first draw of a cleared generator.

One-pole lowpass, state y previous 0 at a trigger:

```
a = exp(-2 * pi * fc / fs)
y[n] = (1 - a) * x[n] + a * y[n-1]
```

Output sum

Instrument level and master are linear gains, not stored in the pattern. Default level 1, default master 1. STAND-IN, they are plain multipliers.

Linear pan, p from -1 to 1:

```
gL = 0.5 * (1 - p)
gR = 0.5 * (1 + p)
```

STAND-IN.

Voice	Pair	p
BD1	BD L	hard left, written only to L at gain 1
BD2	BD R	hard right, written only to R at gain 1
SD	SD/RS L	hard left
RS	SD/RS R	hard right
OH + HH	HH/CY L	summed, hard left
CY	HH/CY R	hard right
CP bursts	CP stereo	pan of burst i, below
CP tail	CP stereo	$p = 0$
LTC	TO/CO	$p = -0.7$
MTC	TO/CO	$p = 0$
HTC	TO/CO	$p = 0.7$
CL	CB/CL L	hard left
CB	CB/CL R	hard right
MA, lead, bass	main only	$p = 0$

The voice sample is after the accent or velocity gain and after the instrument level. Hard left adds that sample to the left channel of the pair and of the main, and adds 0 to the right. Hard right is the swap. A panned voice uses gL and gR on both the pair and the main. The master multiplies the main only. Pair jacks are before the master.

CHOICE: the main mix is that sum whether or not a pair jack is patched. Maracas, lead, and bass have no pair.

Flam table

Flam exists on every drum track except clap. Sixteen patterns, index k from 0 to 15. The manuals name 16 patterns and do not print the timings. STAND-IN offsets, in samples, from the step:

```
hits = 2 + (k mod 4)
gap = 180 * (1 + floor(k / 4))
```

offsets = 0, gap, 2*gap, ... while the list length is hits

Index 0 is samples 0 and 180. Index 15 is 0, 720, 1,440, 2,160, and 2,880. Each offset fires the voice again. A new fire restarts that voice's envelopes. Offsets may cross into the next step.

BD1

```
trig --> pitch envelope --> sine body ----+
      --> transient[s], 4 ms -----+--> sum --> tanh clip --> gain
      --> noise --> one-pole lowpass -----+
```

Decay does not reach a steady tone. The body envelope falls to 0 for every knob value.

```
f_tune = 35 * (140/35) ^ u(tune)          # 35 Hz to 140 Hz
depth  = 18 * u(pitch)                    # semitones, 0 to 18
tau_p  = 0.012 + 0.25 * u(pitch)          # seconds
tau_b  = 0.03 + 1.2 * u(decay)             # seconds, always finite
s      = min(15, floor(trigger_cc * 16 / 128))
f_tr   = 160 * (1.35 ^ s)
tau_tr = 0.004
fc     = 200 * (8000/200) ^ u(filter)     # 200 Hz to 8,000 Hz
drive  = 1 + 8 * u(dist)

p_env[n] = exp(-n / (fs * tau_p))
f[n]     = f_tune * 2 ^ (((depth + bend_st) * p_env[n]) / 12)
body[n]  = sin(phase[n]) * exp(-n / (fs * tau_b))
tr[n]    = sin(2 * pi * f_tr * n / fs) * exp(-n / (fs * tau_tr))
noise    = u(noise_cc) * lowpass(x[n], fc)
pre[n]   = body[n] + u(attack) * tr[n] + noise
y[n]     = tanh(drive * pre[n]) / tanh(drive)
```

phase advances by $2 * \pi * f[n] / fs$ after the sample is taken. n is samples since the trigger. bend_st is the step bend in semitones. It decays with p_env, on the same time constant as the Pitch knob. INT step bend is $bend_st = 12 * (2 * u(bend_cc) - 1)$. CC 0 is -12 semitones, CC 127 is +12, and CC 64 is $12 * (128/127 - 1) = 0.094488$ semitones. STAND-IN. EXT forces bend_st to 0. The printed BD1 example uses bend_st = 0, noise CC 0, and dist CC 0, so y = pre and the noise term is 0.

Example knobs: Attack 64, Decay 80, Pitch 40, Tune 50, Noise 0, Filter 64, Dist 0, Trigger 0.

```
u(attack) = 0.503937
f_tune    = 60.408707 Hz
depth     = 5.669291 semitones
tau_p     = 0.090740 s
tau_b     = 0.785906 s
s         = 0
f_tr      = 160 Hz
```

n	f Hz	y
0	83.814360	0.000000
10	83.751440	0.208884
48	83.514083	0.832547
480	80.998689	-0.907532
2,400	72.957296	-0.605374

The instantaneous frequency at n = 2,400 is closer to f_tune than the frequency at n = 0. That is the pitch envelope decaying.

Same knobs, Attack 127, sample n = 10: Trigger 0 gives s = 0, f_tr = 160, y = 0.306787. Trigger 64 gives s = 8, f_tr = 1,765.184603, y = 0.810530. The attack sample changed because the transient frequency changed.

BD2

```
trig --> sine body, envelope toward sustain or toward 0 --> out
      --> octave sine, 5 ms, level = tone ----->
```

No noise, no filter, no clip, no 16-type transient. The body can hold.

```
f_tune = 45 * (100/45) ^ u(tune)          # 45 Hz to 100 Hz
f_tr   = 2 * f_tune
tau_tr = 0.005

if decay_cc >= 127:
    sustain = 0.70
    tau_b   = 0.40
else:
    sustain = 0.0
    tau_b   = 0.04 + 1.6 * u(decay)

env[n] = sustain + (1 - sustain) * exp(-n / (fs * tau_b))
body[n] = sin(phase[n]) * env[n]
tr[n]   = sin(2 * pi * f_tr * n / fs) * exp(-n / (fs * tau_tr))
y[n]    = body[n] + u(tone) * tr[n]
```

Pitch bend, when the step has one, uses the BD1 pitch-envelope shape with $\tau_p = 0.08$ s STAND-IN and $\text{depth_st} = 12 * (2 * u(\text{bend_cc}) - 1)$. The Tune knob is the resting frequency. The example below has no bend.

Hold check, envelope only: Decay 127 at $n = 0$ is 1.000000, at $n = 480$ is 0.992593, at $n = 96,000$ (2 s) is 0.702021, against sustain 0.700000. Decay 0 at $n = 96,000$ is 0.000000 within $1e-5$. Decay 126 at $n = 96,000$ is 0.292599 and is not a hold. Only CC 127 holds.

Tone example, Tune 60, Tone 100, $n = 10$, envelopes of the transient only: $f_tune = 65.621948$ Hz, $f_tr = 131.243897$ Hz, $u(\text{tone}) * \tau = 0.129116$.

SD

```
trig --> pitch envelope --> sine tone 1 --+
                                --> sine tone 2 --+--> blend --> out
                                --> noise envelope ----->

f1      = 120 * (400/120) ^ u(tune)          # 120 Hz to 400 Hz
detune_st = -8 + 16 * u(d_tune)              # -8 to +8 semitones
f2      = f1 * 2 ^ (detune_st / 12)
depth   = 14 * u(pitch)                     # semitones
tau_p   = 0.01 + 0.12 * u(pitch)
tau_tone = 0.02 + 0.55 * u(tone_decay)
tau_n   = 0.01 + 0.25 * u(sn_decay)

f1[n] = f1 * 2 ^ (depth * p_env[n] / 12)
f2[n] = f2 * 2 ^ (depth * p_env[n] / 12)
t1[n] = sin(phase1) * exp(-n / (fs * tau_tone))
t2[n] = sin(phase2) * exp(-n / (fs * tau_tone))
nz[n] = u(snappy) * x[n] * exp(-n / (fs * tau_n))
y[n]  = (1 - u(tone)) * t1[n] + u(tone) * t2[n] + nz[n]
```

Example, noise left out (Snappy 0): Tune 70, D-Tune 90, Snappy 0, SN Decay 50, Tone 64, Tone Decay 60, Pitch 30.

```
f1      = 233.014061 Hz
f2      = 282.574688 Hz
detune_st = 3.338583 semitones
tau_tone = 0.279843 s
blend    = 0.503937
```

At $n = 20$, $t1 = 0.671594$, $t2 = 0.778808$, $y = 0.725623$. Both tones are in the sum. Tone CC 0 would output $t1$. Tone CC 127 would output $t2$.

RS

```
trig --> short sine --> out

f   = 250 * (2500/250) ^ u(tune)          # 250 Hz to 2,500 Hz
tau = 0.012                               # fixed, no decay knob
```

```
y[n] = sin(2 * pi * f * n / fs) * exp(-n / (fs * tau))
```

Tune 40: f = 516.298233 Hz. At n = 8, y = 0.507608.

CY

```
trig --> metallic stack A --+
      --> metallic stack B --+--> blend --> decay --> out
      --> noise, fixed mix -->
```

Two stacks. Each partial is a sine plus its third harmonic when 3f is under the Nyquist rate, which is the first two terms of a square. Ratios are STAND-IN.

```
ratios_a = 1.00, 1.52, 1.87, 2.41
ratios_b = 1.00, 1.34, 1.71, 2.05
f0 = 180 * (900/180) ^ u(tune)          # 180 Hz to 900 Hz
tau = 0.05 + 1.8 * u(decay)
```

For each ratio r, one partial is $\sin(2 * \pi * f_0 * r * n / fs)$. When $3 * f_0 * r$ is under the Nyquist rate, add one third of the sine at $3 * f_0 * r$. Divide that partial by its weight: 1, or 1 + 1/3 when the third is present. The stack sample is the mean of the four partials. Then:

```
y[n] = exp(-n / (fs * tau)) * (
    (1 - u(tone)) * stack_a[n]
    + u(tone) * stack_b[n]
    + 0.15 * u(tone) * x[n] )
```

The 0.15 noise mix is STAND-IN. The numeric row below uses the stacks only, noise omitted, so the row does not depend on generator position.

Tune 64, Tone 70, Decay 90: f0 = 405.050673 Hz, tau = 1.325591 s, blend = 0.551181. At n = 30, stack A = 0.187710, stack B = 0.382957, y = 0.295187.

OH and HH

One colour, two decays, one choke.

```
f_c    = 250 * (4000/250) ^ u(hh_tune)    # 250 Hz to 4,000 Hz, CC 73
tau_oh = 0.04 + 1.1 * u(oh_decay)
tau_hh = 0.008 + 0.04 * u(hh_decay)
```

Each hat is a metallic stack at f_c with ratios 1.00, 1.47, 1.80, 2.33, built like a CY stack, plus noise at mix $0.25 * x[n]$, times its own exp decay. STAND-IN ratios.

A closed-hat trigger sets the open-hat envelope to 0 on that sample, and the open hat stays at 0 until the next open trigger. It does not shorten the closed hat.

Example colour, HH Tune 60: f_c = 926.436359 Hz. OH Decay 100: tau_oh = 0.906142 s. HH Decay 40: tau_hh = 0.020598 s. An open hat that has run 200 samples has envelope 0.995412. On the closed trigger that envelope is 0.

CL

```
f    = 400 * (3000/400) ^ u(tune)          # 400 Hz to 3,000 Hz
tau  = 0.004 + 0.08 * u(decay)
y[n] = sin(2 * pi * f * n / fs) * exp(-n / (fs * tau))
```

Tune 50, Decay 30: f = 884.244363 Hz, tau = 0.022898 s. At n = 12, y = 0.972835.

CP

```
trig --> N short sines, 11 ms apart, panned --> stereo
      --> filtered noise tail -----> both channels

s      = min(15, floor(trigger_cc * 16 / 128))    # CC 77, ASSUMED type
count  = 1 + floor(data_cc * 8 / 128)             # 1 to 8, ASSUMED, no CC
```

```

gap      = round(0.011 * fs)                # 528 samples, 11 ms
f_tr     = 700 * (1.28 ^ s)
tau_tr   = 0.003
fc       = 400 * (6000/400) ^ u(filter)
tau      = 0.05 + 0.8 * u(decay)

```

Burst i , i from 0 to count-1, starts at sample $i * \text{gap}$. Its own $n_b = n - i * \text{gap}$, and it is 0 when $n_b < 0$.

```
burst_i[n] = sin(2 * pi * f_tr * n_b / fs) * exp(-n_b / (fs * tau_tr))
```

Pan of burst i , $\text{count} > 1$: $p = -1 + 2 * i / (\text{count} - 1)$. One burst uses $p = 0$. The tail is noise through the one-pole at f_c , times $\exp(-n / (fs * \tau))$, times 0.5 on each channel. The tail is the manual's reverb tail rendered as a noise decay. It is not a room.

```

yL[n] = u(attack) * sum_i burst_i[n] * gL(p_i) + tail[n] * 0.5
yR[n] = u(attack) * sum_i burst_i[n] * gR(p_i) + tail[n] * 0.5

```

Data CC 48 gives count 4. gap is 528. With Trigger 0, $f_{tr} = 700$ Hz. The transient at 5 samples after each burst start, before Attack scaling, is 0.427195. Burst starts are samples 0, 528, 1,056, and 1,584.

LTC, MTC, HTC

Each voice:

```

trig --> sine body, BD2-style hold --> out
      --> noise, if that voice's switch is on, shared level

```

Tom mode is one sine at f_{tune} . Conga mode adds a second sine at $2.30 * f_{tune}$, gain 0.35. ASSUMED. The manuals say the switch changes tom and conga and do not describe the waveform.

```

LTC f = 70 * (180/70) ^ u(tune)
MTC f = 100 * (280/100) ^ u(tune)
HTC f = 140 * (400/140) ^ u(tune)

```

Decay matches the BD2 hold law with different constants, so a tom is not a copy of BD2. STAND-IN:

```

if decay_cc >= 127:
    sustain = 0.55
    tau_b   = 0.35
else:
    sustain = 0
    tau_b   = 0.03 + 1.2 * u(decay)

```

Noise, when the voice switch is on (CC at or above 64): $u(\text{tom_noise}) * x[n] * \exp(-n / (fs * 0.12))$. The 0.12 s noise decay is STAND-IN and shared. tom_noise is CC 84.

LTC Tune 40: $f = 94.251191$ Hz. Decay 127 at $n = 96,000$: envelope 0.551484, sustain 0.550000. TOM_NOISE 33: level 0.259843.

Bend uses the BD2 bend paragraph.

CB

```

trig --> square at f ----+
      --> square at 1.015 f --> average --> decay --> out

```

The two squares are the brief. Detune ratio 1.015 is STAND-IN. A square is the odd harmonics $k = 1, 3, 5, \dots$ while $k * f$ is under Nyquist and $k \leq 15$, amplitude $1/k$, divided by the sum of those amplitudes.

```

f      = 300 * (1200/300) ^ u(tune)          # 300 Hz to 1,200 Hz
tau    = 0.01 + 0.40 * u(decay)
y[n]   = 0.5 * (square(f) + square(1.015 * f)) * exp(-n / (fs * tau))

```

Tune 48, Decay 70: $f = 506.607352$ Hz, second square 514.206462 Hz, $\tau = 0.230472$ s. At $n = 15$, $y = 0.405275$.

Sixteen panel steps, if a UI wants them, are Tune CC values $\text{round}(i * 127 / 15)$ for i from 0 to 15. The voice reads the continuous CC.

MA

```
trig --> noise --> one-pole lowpass at 1,500 Hz --> decay --> main
```

No tune knob. Colour is fixed. STAND-IN $f_c = 1,500$ Hz. The filter is the lowpass in the shared section, held at that cutoff.

Decay:

```
tau = 0.02 + 0.30 * u(decay)
y[n] = lowpass(x[n]) * exp(-n / (fs * tau))
```

Decay 55: $\tau = 0.149921$ s. First noise draw $x = -0.527089$. $a = \exp(-2 * \pi * 1500 / fs) = 0.821725$. With filter state 0, $y[0] = -0.093967$ before the envelope, and the envelope at $n = 0$ is 1.

Lead and bass

```
note --> band-limited saw --> one-pole lowpass --> envelope --> main
```

Waveform ASSUMED: saw, harmonics $k = 1$ to 8, amplitude $1/k$, skipped when $k * f$ is at or above Nyquist, then divided by the sum of the weights used.

Pitch is MIDI note 36 to 72, which is the appendix range for CV1 and CV23. $\text{Hz} = 440 * 2^{((\text{note} - 69) / 12)}$. Note 60 is 261.625565 Hz. Note 36 is 65.406391 Hz. Note 72 is 523.251131 Hz.

The panel entry in both manuals is step buttons 1 to 13 for pitch class C through the next C, and buttons 14 to 16 for the octave. ASSUMED map onto notes 36 to 72: octave 0, 1, 2 from buttons 14, 15, 16, pitch class 0 to 12 from buttons 1 to 13, $\text{note} = 36 + 12 * \text{octave} + \text{class}$, then clamp to 72. A class of 12 is the upper C, so some combinations land on the same note. That collision is accepted.

A silent step, the A/B rest, forces the voice to 0 for the samples of that step, after the release below. A tied note keeps the envelope open and does not retrigger.

Envelope: attack is one sample to 1 on a new note, hold while the note is tied, release $\tau = 0.03$ s STAND-IN when the note ends. A rest is not a release of a previous note inside this test's cleared state. In play, a rest after a note starts the release.

Lead tone, ASSUMED, one-pole lowpass:

```
fc = 200 * (8000/200) ^ u(lead_tone)
```

Bass tone is the manual's cutoff, and the same u is the CV3 value, scaled STAND-IN as 0 V to 5 V for a jack: $\text{cv3} = 5 * u(\text{bass_tone})$.

```
fc = 80 * (4000/80) ^ u(bass_tone)
```

Lead example, note 60, before the filter, $n = 25$ samples after a trigger, harmonics 1 to 8: $\text{saw} = 0.387776$. Lead tone CC 80: $f_c = 2,042.686148$ Hz.

Bass accent uses g_accent on the bass voice in INT. The drums' three-level accent is the same table.

Step bend and the Pitch knob

BD1's Pitch knob is both time and depth, one u , as the instrument chapter states. SD's pitch control is depth, with the time law above. Step bend is an extra signed depth on BD1, BD2, SD, LTC, MTC, and HTC only. CY, RS, hats, clave, clap, cowbell, and maracas have no step bend. EXT ignores step bend.

Cleared voice

A voice that has never been triggered in the block under test outputs 0 on every sample. A drum step that is off does not trigger. A note-track rest does not open the gate. Neither event produces a new nonzero sample from a cleared voice.

Build guide (2026-10-06)

This commit builds the voice engine and a PDF. It does not build a plugin.

This tree

The voice engine is `engine/`. Panel art is unpainted. JIDAI RACK is not changed by this repository. The equations in `SCHEMATICS.md` are the contract. `make test` builds `build/shogun_tests` and runs the checks in `TESTPLAN.md`.

Rebuild the pack PDF from the markdown:

```
python3 scripts/build_design_pack_pdf.py
```

The script reads `README.md`, `METHODOLOGY.md`, `SCHEMATICS.md`, `BUILD_GUIDE.md`, `TESTPLAN.md`, and `REPO_SETUP.md`, in that order, and writes `docs/SHOGUN_Design_Pack.pdf`. It needs `reportlab`. It does not fetch audio code.

Engine

Piece	Job
Clock	INT and EXT, tempo, scale, the step counter that keeps running in EXT
Pattern	14 drum tracks, 2 note tracks, lengths, shuffle, shift, mute, per-step flags
Voices	One block per voice in <code>SCHEMATICS.md</code> , cleared to 0 until triggered
Outs	The pair map, the main sum, master after the sum
Tests	The names in <code>TESTPLAN.md</code> , against the printed 48 kHz rows

Host tempo wins while the host is playing. Otherwise the internal tempo, 60 to 180 BPM, default 120, is the clock. The display shows BPM and the step. EXT leaves both of those moving.

Trig jacks are rising edges through 1.0 V. In INT they are not read. In EXT they are the only triggers. Velocity on the edge is the accent in EXT. A gate with no velocity byte uses 127.

Live knobs apply in both modes. Pattern step on, flam, accent, and step bend apply only in INT.

Panel

The jobs may match the instrument chapter: the same knob names, the same 14 drums, the two note tracks, the INT/EXT switch, and a Trig jack on each voice. The face must not be a trace of the hardware panel. No MFB logo. No dancing-bear mark. Draw a new plate. Level and master stay off the pattern, as in both manuals.

The INT/EXT switch is a SHOGUN control. Putting it on the plate does not make it a copy of a Tanzbär control, because that machine does not have this switch.

Outputs

Pairs, post level and post accent, before master:

- BD: left BD1, right BD2
- SD/RS: left snare, right rim
- HH/CY: left open plus closed, right cymbal
- CP: stereo clap
- TO/CO: LTC, MTC, and HTC at pans -0.7, 0, and 0.7
- CB/CL: left clave, right cowbell

Main is the sum of those contributions, plus maracas, lead, and bass at center pan, then the master. A patched pair does not remove its voice from the main.

Limits

The voice is the equation set in SCHEMATICS.md. A STAND-IN number stays marked until a measurement replaces it. A sampled kit is outside this design. A transistor netlist is outside this design: the manuals did not provide one. JIDAI RACK does not compile this repository, and this guide does not add it.

Test plan (2026-10-06)

These tests are the contract for the engine in engine/. tests/voices.cpp compiles them into build/shogun_tests. Each one is specified against SCHEMATICS.md at 48,000 Hz, double precision, absolute tolerance 1e-5 on the printed decimals unless a row says otherwise.

A cleared voice is one that has not been triggered. Its output block is 0.

Shared pattern for the bypass tests: 16th scale, 120 BPM, so a step is 6,000 samples. Pattern length 4. Track length 4. One drum voice, BD1, with the BD1 example knobs (Attack 64, Decay 80, Pitch 40, Tune 50, Noise 0, Filter 64, Dist 0, Trigger 0). Step 0 is on. Steps 1, 2, and 3 are off. No flam. Accent index 2. No bend.

testKickBendDecays

Render BD1 from the example knobs, bend_st 0, from a trigger at $n = 0$.

The frequency at $n = 0$ is 83.814360 Hz. The frequency at $n = 2,400$ is 72.957296 Hz. f_{tune} is 60.408707 Hz. The later frequency is closer to f_{tune} than the first. y at $n = 48$ is 0.832547. y at $n = 480$ is -0.907532.

Fail if the later frequency is farther from f_{tune} than the frequency at the trigger, or if either printed y misses by more than 1e-5.

testBd1SoundChangesAttack

Same knobs, Attack raised to 127. Render two triggers.

Trigger CC 0: at $n = 10$, $y = 0.306787$, transient frequency 160 Hz.

Trigger CC 64: at $n = 10$, $y = 0.810530$, transient frequency 1,765.184603 Hz.

The two y values differ. The body tune is the same in both renders, so the difference is the attack transient.

testBd2CanHold

BD2 envelope, no dependence on phase.

Decay CC 127 at $n = 96,000$ is 0.702021, and the sustain it is approaching is 0.700000. The absolute difference from 0.70 is under 0.01.

Decay CC 0 at $n = 96,000$ is 0 within 1e-5.

Decay CC 126 at $n = 96,000$ is 0.292599. That value is not a hold. The test fails if CC 126 is treated as a hold, or if CC 127 has fallen below 0.5.

Tune 60 and Tone 100, transient only, at $n = 10$: scaled transient 0.129116. This row keeps the tone path from being a silent stub.

testSnareTwoTones

SD example knobs, Snappy 0 so the noise term is 0. Tune 70, D-Tune 90, Tone 64, Tone Decay 60, Pitch 30, SN Decay 50.

f1 is 233.014061 Hz. f2 is 282.574688 Hz. At $n = 20$, t_1 is 0.671594, t_2 is 0.778808, and y is 0.725623.

A second render with Tone CC 0 equals t_1 at $n = 20$. A third with Tone CC 127 equals t_2 at $n = 20$. Both partials are present at Tone 64, and the blend knob selects between them.

testHatChoke

HH Tune 60, OH Decay 100, HH Decay 40. Open hat triggers at sample 0. Envelope at sample 200, before any closed trigger, is 0.995412.

A closed-hat trigger at sample 200 sets the open envelope to 0 on that sample. Samples 200 through 500 of the open hat are 0. A later open trigger may sound again. The closed hat's own tau is 0.020598 s and is not cleared by its own trigger.

testClapBurstCount

Data CC 48, so count is 4. Trigger CC 0, so the type index is 0 and f_{tr} is 700 Hz. Gap is 528 samples.

Four bursts start at samples 0, 528, 1,056, and 1,584. The gap in time is 11 ms, which is inside 10 to 12 ms. The unscaled transient at 5 samples after each start is 0.427195.

A count of 1 produces one start and no second peak 528 samples later. The voice is not a single noise hit when count is 4: four starts exist, and they are not on the same sample.

Flam is not applied. The manuals exclude clap from flam, and this test does not put a flam on it.

testExtBypassIgnoresPattern

Mode EXT. The shared pattern has step 0 on. No Trig edge in the first step.

The BD1 output over the first 6,000 samples is 0. The clock counter still advances: after 6,000 samples the display step has moved.

Then a Trig rising edge, velocity 100, at the start of the next block, with the pattern still showing an off step or an on step. The voice fires. $g_{vel}(100)$ is 0.819291. y at 10 samples after that edge equals the BD1 example y at $n = 10$, which is 0.208884, multiplied by 0.819291.

The pattern accent is index 2, whose g_{accent} is 1. The EXT render uses 0.819291, not 1. A flam on the pattern step does not add the extra hits at 180 samples.

Switch to INT on a later step without resetting the counter. The next on-step of the pattern fires, and the Trig jack no longer fires.

testIntIgnoresTrigJacks

Mode INT. Same pattern. A Trig rising edge is placed in the middle of step 1, which is off. The voice stays 0 through that step.

Step 0, with no dependence on the Trig jack, fires. With accent index 2, y at 10 samples after the step boundary is 0.208884 times 1.000000.

Pulling the Trig cable, or leaving it plugged, does not change INT. The switch does.

testRestIsSilent

From cleared voices:

- A drum step that is off, one full step at the clock above, outputs 0.
- A lead note-track rest, note slot empty, outputs 0 for that step.
- A bass note-track rest outputs 0 for that step.

A previous hit is not part of this test. The rest does not have to choke a voice that is already ringing. Hats have their own choke test.

testIndividualOutStaysInMix

One BD1 hit, level 1, master 1, accent gain 1, the example hit. The BD pair's left channel at $n = 48$ is 0.832547. The main left channel at $n = 48$ is the same 0.832547. The pair is treated as patched.

Repeat with the pair unpatched. Main left at $n = 48$ is still 0.832547.

Master 0.5 scales the main to 0.416273 and leaves the pair at 0.832547.

Maracas does not appear on any pair. A maracas trigger with the first noise draw appears on the main only.

Also locked by those tests

- EXT does not freeze the counter: testExtBypassIgnoresPattern reads the display after one silent step.
- A Trig cable does not select EXT: testIntIgnoresTrigJacks keeps the cable connected.
- Clap spacing is 11 ms: testClapBurstCount.
- BD2 hold is CC 127 only: testBd2CanHold.

Repository setup (2026-10-06)

Tree

Public repository: <https://github.com/martialsystems/shogun>

Working name: SHOGUN. A separate product name has not been set. The engine commit is on branch feat/engine-voices, cut from the design pack in this repository.

This commit contains:

Path	Role
README.md	Front page and revision list
METHODOLOGY.md	Sources and the quoted choices
SCHEMATICS.md	Equations and 48 kHz rows
BUILD_GUIDE.md	PDF build, and the engine layout
TESTPLAN.md	Named tests
REPO_SETUP.md	This file
engine/dsp.h	Shared STAND-IN helpers
engine/shogun.h	Clock, pattern, knobs, and the process call
engine/shogun.cpp	Voices and the output sum
tests/voices.cpp	Named checks
Makefile	Builds build/shogun_tests
scripts/build_design_pack_pdf.py	Markdown to PDF
docs/SHOGUN_Design_Pack.pdf	The pack as one PDF
LICENSE	All rights reserved, Martial Systems LLC
.gitignore	build/ and __pycache__/

The voice engine is framework-free C++ under engine/. process() does not allocate. There is no panel SVG and no JUCE target.

The MFB manuals are not in the tree. The German PDF that was read for the quotations stays outside the repository. The pack cites it. It does not redistribute it.

PDF

```
python3 scripts/build_design_pack_pdf.py
```

The script fails if a source file contains an em dash, an en dash, or a heading that defines the work by absence. The PDF is a flowing render of the six markdown files, in the order README, METHODOLOGY, SCHEMATICS, BUILD_GUIDE, TESTPLAN, REPO_SETUP. A title line on each PDF page carries the date 2026-10-06. That date is the revision date. A clock time inside the PDF metadata is only the build time.

Collection link

martialsystems/jidai-collection names this repository under the SHOGUN heading on main. The rack plugin in that repository compiles BUSHIDO and RONIN. It does not fetch SHOGUN. This engine does not change the rack.

Git

Commits on this repository use author Martial Systems LLC and 25778085+martialsystems@users.noreply.github.com, the same author as the collection repository.

The license grants no right to copy the pack without permission from Martial Systems LLC. Short quotations in METHODOLOGY.md are the disputed parameter sentences, with the source named beside each one.

Clock switch, as a repository fact

INT and EXT are specified in METHODOLOGY.md and SCHEMATICS.md. They are not a control from the Tanzbär manuals. A later patch against this repository that removes the switch, or that lets a Trig cable force EXT, breaks testExtBypassIgnoresPattern and testIntIgnoresTrigJacks.